

SMishReLU: A Continuous C^1 -Smooth Hybrid Activation Function Eliminating First-Derivative Discontinuity in Deep Neural Networks

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Abstract

Activation functions play a foundational role in controlling non-linearity, gradient flow, and optimization dynamics in deep neural networks (DNNs). Recently, hybrid activation functions such as MishReLU were introduced to address the "dying ReLU" problem by combining the positive linearity of ReLU with the smooth negative concavity of Mish. However, a rigorous mathematical examination reveals that MishReLU suffers from an unaddressed first-derivative jump discontinuity at $x = 0$, where its left-hand derivative ($\lim_{x \rightarrow 0^-} \text{MishReLU}'(x) = \tanh(\ln 2) \approx 0.6000$) fails to match its right-hand derivative ($\lim_{x \rightarrow 0^+} \text{MishReLU}'(x) = 1.0000$). This abrupt 0.4000 derivative step induces gradient shocks during backpropagation, destabilizing optimization in deep residual and recurrent architectures. To overcome this limitation, we propose SMishReLU (*Smooth Mish-ReLU*), a continuously differentiable (C^1 -smooth) hybrid activation function. By scaling the negative branch with a mathematically derived coefficient $\alpha = \frac{1}{\tanh(\ln 2)} \approx 1.6667$, SMishReLU forces both the left-hand and right-hand derivatives at $x = 0$ to equal exactly 1.0000. Extensive empirical evaluations across five benchmark datasets—spanning image classification (MNIST, Fashion-MNIST, CIFAR-100) and text classification (IMDB, Reuters) across MLP, CNN, ResNet-18, LSTM, and BiLSTM architectures—demonstrate that SMishReLU consistently outperforms MishReLU and standard baselines in accuracy, convergence speed, and gradient stability.

Keywords— Activation functions, Deep neural networks, Gradient shock, Derivative continuity, MishReLU, C^1 smoothness, Literature review, Deep learning optimization.

1 Introduction

Deep neural networks (DNNs) owe their expressive power to non-linear activation functions, which allow complex hierarchical abstractions to be learned from high-dimensional data. Standard functions such as Rectified Linear Unit (ReLU) [2] gained widespread popularity due to their computational simplicity and ability to mitigate the vanishing gradient problem in the positive domain. However, ReLU sets all negative activations to zero ($\text{ReLU}(x) = 0$ for $x \leq 0$), leading to the "dying ReLU" phenomenon where neurons receive zero gradients and become permanently inactive during training.

To overcome the dead neuron issue, various non-zero negative activation functions have been developed, including LeakyReLU [3], Exponential Linear Unit (ELU) [4], Scaled Exponential Linear Unit (SELU) [5], Swish [6], and Mish [7]. Recently, hybrid activation functions have gained attention for combining the strengths of distinct functional profiles. In particular, the **MishReLU** activation function [1] was introduced to unify the strict identity mapping of ReLU for positive inputs ($x > 0$) with the smooth non-monotonic well of Mish ($x \cdot \tanh(\text{softplus}(x))$) for negative inputs ($x \leq 0$).

Despite its empirical success, our theoretical analysis reveals a critical mathematical flaw in MishReLU: **a first-derivative jump discontinuity at $x = 0$** . Specifically, as x approaches 0 from the negative domain, the first derivative evaluates to $\tanh(\ln 2) \approx 0.6000$, whereas from the positive domain, it equals 1.0000. This sudden step change of $\Delta g = 0.4000$ violates C^1 differentiability across the origin, creating gradient shocks whenever neuron pre-activations oscillate across zero during backpropagation.

In this work, we present **SMishReLU** (*Smooth Mish-ReLU*), a novel C^1 -continuous hybrid activation function that resolves the derivative jump discontinuity of MishReLU. By introducing an exact scaling coefficient $\alpha = \frac{1}{\tanh(\ln 2)} = \frac{5}{3} \approx 1.6667$ into the negative branch, SMishReLU guarantees equal left- and right-hand derivatives at $x = 0$, ensuring seamless gradient flow.

The key contributions of this paper are summarized as follows:

1. **Mathematical Analysis:** We rigorously identify and prove the C^1 derivative jump discontinuity of MishReLU at $x = 0$.
2. **Proposed SMishReLU Function:** We formulate SMishReLU and provide a formal proof of its complete C^1 differentiability across $\mathbb{R}$.
3. **Extensive Literature Review:** We review 16 recent activation function publications (2020–2025), positioning our contribution within the state-of-the-art.
4. **Implementation & Visualization:** We provide lightweight implementations in TensorFlow/Keras and PyTorch, along with detailed curve and derivative visualizations.
5. **Comprehensive Empirical Validation:** We replicate the experimental framework of the baseline paper across 5 benchmark datasets (MNIST, Fashion-MNIST, CIFAR-100, IMDB, Reuters) over 5 random seeds, demonstrating superior accuracy and stability.

2 Literature Review of Recent Activation Functions (2020–2025)

Activation function design continues to be one of the most active subfields in deep learning optimization. Researchers continuously propose novel formulations to strike an optimal balance among gradient flow, non-monotonicity, computational overhead, and derivative smoothness. In this section, we review 16 recent activation functions published between 2020 and 2025, categorizing their functional characteristics and identifying their structural limitations.

2.1 Survey and Comprehensive Review Papers

Several recent survey papers systematically summarize the progression of activation functions:

- **Nwankpa et al. (2018) [8]:** Provided an early comprehensive survey comparing traditional activations (Sigmoid, Tanh, ReLU) with emerging variants, emphasizing their role in feature identification and pattern recognition.
- **Dubey et al. (2022) [9]:** Conducted an extensive survey categorized by exponential, piece-wise linear, and adaptive formulations, concluding that smooth non-monotonic functions (such as Swish, Mish, and PAU) perform best in speech and vision tasks.
- **Rasamoelina et al. (2020) [10]:** Explored automated search methods for function discovery and

analyzed how smooth activations simplify optimization landscapes.

- **Szandała (2021) [11]:** Evaluated commonly used activation functions across deep neural network architectures, demonstrating that no single activation function dominates all datasets.
- **Pusztaházi et al. (2024) [12]:** Provided a detailed tutorial survey focusing specifically on parametric activation functions, detailing the trade-offs between fixed parameters and learnable coefficients.
- **Emanuel et al. (2024) [13]:** Analyzed the specific impact of activation functions on CNN text classification models, measuring convergence speed, accuracy, and misclassification confidence.

2.2 Recent Novel Activation Functions (2020–2025)

In addition to surveys, numerous novel functions have been proposed to address specific optimization challenges:

1. **ActFun (2025) [14]:** Formulated as $\frac{1}{\beta} \log(1 + e^{\beta x})$, ActFun provides a smooth, monotonic, continuously differentiable activation function that avoids gradient saturation.
2. **Cauchy Activation (Li et al., 2025) [15]:** Defined as $f(x) = \lambda_1 \frac{x}{x^2 + d^2} + \lambda_2 \frac{1}{x^2 + d^2}$, this function incorporates Cauchy distribution characteristics to achieve a non-monotonic, zero-centered, double-bounded profile for robust out-of-distribution handling.
3. **Dynamic Activation for OOD (Ji et al., 2025) [16]:** Introduced a dynamic activation mechanism in ICASSP 2025 to enhance out-of-distribution detection capabilities in deep classifiers.
4. **Greenhouse Custom Activation (Airlangga et al., 2025) [17]:** Designed a custom activation function combined with a custom loss for hybrid CNN-LSTM time-series models in agricultural prediction.
5. **APALU (Subramanian et al., 2024) [18]:** Formulated a trainable, adaptive activation function combining a modified linear term for positive inputs $a \left(x + \frac{x}{1 + e^{-1.702x}} \right)$ with an exponential decay for negative inputs $b(e^x - 1)$.
6. **OP-Tanish (Ankalaki & Thippeswamy, 2024) [19]:** Defined as $x \cdot \tanh \left(\frac{x}{1 + e^{-\beta x}} \right)$, providing a smooth, non-monotonic, unbounded curve for 1D-CNN sensor feature extraction.
7. **Trish (Közkurt et al., 2024) [20]:** Formulated as $x \cdot \text{sigmoid}(\ln(1 + \tanh(x)))$, Trish offers an efficient non-monotonic profile optimized for medical image classification and glaucoma diagnosis.

8. **ModExp & ModExp_m** (Kalim et al., 2024) [21]: Defined as $\frac{x}{x^2+e^{-x}}$ and $\frac{x}{x^2+e^x}$, these smooth, continuously differentiable, bounded functions enhance learning capacity without exploding gradients.
9. **NIPUNA** (Madhu et al., 2023) [22]: Defined as $\max\left(\frac{x}{1+e^{-\beta x}}, x\right)$, combining self-gated rectification with smooth negative concavity to mitigate neuron dying.
10. **TaLU** (Hasan et al., 2023) [23]: Proposed a hybrid function combining Tanh and ReLU ($\text{ReLU}(x) + \text{Tanh}(x)$) to compensate for information loss in negative activation regions.
11. **Smish** (Wang et al., 2022) [24]: Defined as $x \cdot \tanh(\ln(1 + \sigma(x)))$, providing a smooth lower-bounded non-monotonic profile.
12. **AbsLU & IpLU** (Alkhouly et al., 2021) [25]: AbsLU ($\alpha|x|$) and IpLU ($\frac{x}{1+|x|^\alpha}$) introduced negative region modulation to sustain gradient flow in deep feedforward networks.
13. **RSigELU** (Kılıçarslan et al., 2021) [27]: Formulated a nonlinear parametric ELU variant providing parametric smoothness for $x \geq 0$ and exponential decay for $x < 0$.
14. **TanhEx** (Xinyu, 2020) [28]: Formulated as $x \cdot \tanh(e^x)$, achieving high convergence speed in lightweight neural networks.
15. **RMAF** (Yu et al., 2020) [29]: Combined ReLU with a memristive window function to model nonlinear hardware synapses in deep learning.
16. **MishReLU** (Al-Nawi et al., 2026) [1]: Merged positive identity mapping (x) with negative Mish ($x \cdot \tanh(\text{softplus}(x))$). While effective, MishReLU introduces a derivative jump from 0.6000 to 1.0000 at $x = 0$, which our proposed SMishReLU directly solves.

3 Proposed Methodology

3.1 Mathematical Analysis of MishReLU Discontinuity

The original MishReLU function is defined as:

$$\text{MishReLU}(x) = \begin{cases} x, & \text{if } x > 0 \\ x \cdot \tanh(\ln(1 + e^x)), & \text{if } x \leq 0 \end{cases} \quad (1)$$

To examine its C^1 differentiability, we evaluate the left-hand and right-hand limits of the first derivative $\text{MishReLU}'(x)$ as $x \rightarrow 0$:

1) *Right-hand derivative* ($x \rightarrow 0^+$):

$$\lim_{x \rightarrow 0^+} \text{MishReLU}'(x) = \frac{d}{dx}[x] = 1.0000 \quad (2)$$

2) *Left-hand derivative* ($x \rightarrow 0^-$): For $x \leq 0$, applying the product rule yields:

$$\begin{aligned} \text{MishReLU}'(x) &= \tanh(\text{softplus}(x)) \\ &\quad + x \cdot \text{sech}^2(\text{softplus}(x)) \cdot \sigma(x) \end{aligned} \quad (3)$$

Evaluating at $x = 0^-$:

$$\lim_{x \rightarrow 0^-} \text{MishReLU}'(x) = \tanh(\text{softplus}(0)) + 0 = \tanh(\ln 2) \approx 0.6000 \quad (4)$$

Because $\lim_{x \rightarrow 0^-} \text{MishReLU}'(x) = 0.6000 \neq 1.0000 = \lim_{x \rightarrow 0^+} \text{MishReLU}'(x)$, MishReLU exhibits a discontinuous derivative jump of $\Delta g = 0.4000$ at $x = 0$.

3.2 Formulation of SMishReLU

To overcome this derivative mismatch while preserving the non-monotonic negative well, we propose **SMishReLU**:

$$\text{SMishReLU}(x) = \begin{cases} x, & \text{if } x > 0 \\ \alpha \cdot x \cdot \tanh(\text{softplus}(x)), & \text{if } x \leq 0 \end{cases} \quad (5)$$

where the scaling factor α is derived as the exact reciprocal of $\tanh(\ln 2)$:

$$\alpha = \frac{1}{\tanh(\ln 2)} = \frac{1}{\frac{2-0.5}{2+0.5}} = \frac{1}{0.6} = \frac{5}{3} \approx 1.666667 \quad (6)$$

3.3 Proof of C^1 Differentiability

Theorem 1. *SMishReLU is continuously differentiable (C^1 -smooth) on $\mathbb{R}$.*

Proof. For $x \neq 0$, SMishReLU is a composition of smooth functions and is therefore C^∞ . At $x = 0$:

$$\text{SMishReLU}'(0^+) = 1.0000$$

$$\text{SMishReLU}'(0^-) = \alpha \cdot \tanh(\ln 2) = \left(\frac{1}{\tanh(\ln 2)} \right) \cdot \tanh(\ln 2) = 1.0000$$

Since $\text{SMishReLU}'(0^-) = \text{SMishReLU}'(0^+) = 1.0000$, the first derivative is continuous everywhere on $\mathbb{R}$. $\square$

3.4 Graphical Comparison

Fig. 1 visually highlights the derivative behavior across functions. Unlike MishReLU's discontinuous step at $x = 0$, SMishReLU smoothly transitions into the positive domain with a continuous derivative of 1.0000.

Table 1: Taxonomy and Analytical Properties of 16 Recent Activation Functions (2020–2025)

Activation Function	Mathematical Definition	Year	Smooth (C^1)?	Primary Advantage / Application
ActFun [14]	$\frac{1}{\beta} \log(1 + e^{\beta x})$	2025	Yes	Monotonic smooth activation
Cauchy [15]	$\lambda_1 \frac{x}{x^2+d^2} + \lambda_2 \frac{1}{x^2+d^2}$	2025	Yes	Non-monotonic, zero-centered
Dynamic Activation [16]	Dynamic gating mechanism	2025	Yes	Out-of-distribution (OOD) d
Greenhouse Custom [17]	Hybrid CNN-LSTM formulation	2025	Yes	Time-series actuator state pr
APALU [18]	$a(x + \frac{x}{1+e^{-1.702x}})$ if $x \geq 0$ else $b(e^x - 1)$	2024	Yes	Adaptive piecewise linear tra
OP-Tanish [19]	$x \cdot \tanh\left(\frac{x}{1+e^{-\beta x}}\right)$	2024	Yes	Smooth 1D-CNN sensor anon
Trish [20]	$x \cdot \text{sigmoid}(\ln(1 + \tanh(x)))$	2024	Yes	Efficient non-monotonic CNN
ModExp [21]	$\frac{x}{x^2+e^{-x}}$	2024	Yes	Continuously differentiable b
NIPUNA [22]	$\max\left(\frac{x}{1+e^{-\beta x}}, x\right)$	2023	No (at peak)	Hybrid saturating self-gated
TaLU [23]	$\text{ReLU}(x) + \text{Tanh}(x)$	2023	No (at 0)	Information loss mitigation in
Smish [24]	$x \cdot \tanh(\ln(1 + \sigma(x)))$	2022	Yes	Lower-bounded non-monoton
AbsLU [25]	x if $x \geq 0$ else $\alpha x $	2021	No (at 0)	Absolute value negative hand
IpLU [26]	x if $x \geq 0$ else $\frac{x}{1+ x ^\alpha}$	2021	Yes ($\alpha = 1$)	Inverse parametric non-zero g
RSigELU [27]	Parametric ELU variant	2021	Yes	Non-linear parametric activat
TanhEx [28]	$x \cdot \tanh(e^x)$	2020	Yes	High convergence speed in lig
MishReLU [1]	x if $x > 0$ else $x \cdot \tanh(\text{softplus}(x))$	2026	No (Jump at 0)	Hybrid ReLU-Mish (suffers 0
SMishReLU (Ours)	x if $x > 0$ else 1.6667 · $x \cdot \tanh(\text{softplus}(x))$	2026	Yes (C^1 Smooth)	Eliminates derivative jum

3.5 Implementation

Listing 1: Keras Implementation of SMishReLU

```
import tensorflow as tf
import numpy as np

ALPHA_SMISH = 1.0 / np.tanh(np.log(2.0)) #
1.666667

def SMishReLU(x):
    mish_part = ALPHA_SMISH * x * tf.keras.
        activations.tanh(
            tf.keras.activations.softplus(x))
    return tf.where(x > 0, x, mish_part)
```

smishrelu_vs_mishrelu_comparison.png

Figure 1: Comparison of activation output $f(x)$ (left) and first derivative $f'(x)$ (right) for ReLU, MishReLU, and proposed SMishReLU. MishReLU exhibits a derivative jump at $x = 0$ ($0.6000 \rightarrow 1.0000$), whereas SMishReLU achieves complete C^1 derivative continuity at 1.0000.

4 Experimental Setup

Experiments strictly follow the baseline repository protocol [30]. Evaluation was conducted across five benchmark datasets:

1. **Fashion-MNIST**: 60,000 train / 10,000 test images using a 3-layer MLP.
2. **MNIST**: 60,000 train / 10,000 test digit images using a 3-block CNN.
3. **CIFAR-100**: 50,000 train / 10,000 test images using ResNet-18.
4. **IMDB**: 25,000 train / 25,000 test movie reviews using a single-layer LSTM.
5. **Reuters**: 8,982 train / 2,246 test newswire texts using a BiLSTM.

All models were trained across 5 random seeds to report mean $\pm$ standard deviation across learning rates (0.01, 0.001, 0.0001).

5 Experimental Results and Discussions

5.1 Fashion-MNIST (MLP Architecture)

Table 2 shows results on Fashion-MNIST. SMishReLU achieves a top accuracy of $89.12\% \pm 0.0028$ at $LR = 0.0001$, surpassing MishReLU (88.69%) and standard Mish (88.61%).

Table 2: MLP on Fashion-MNIST Dataset (Mean $\pm$ SD over 5 seeds)

Metric (%)	LR	SMishReLU	MishReLU	ReLU	Mish	ELU	LeakyReLU	SeLU
Accuracy	0.01	87.85 $\pm$ 0.0031	86.97 $\pm$ 0.0072	87.00 $\pm$ 0.0033	87.06 $\pm$ 0.0054	87.21 $\pm$ 0.0060	87.17 $\pm$ 0.0055	87.52 $\pm$ 0.0059
	0.001	89.04 $\pm$ 0.0029	88.65 $\pm$ 0.0057	87.43 $\pm$ 0.0067	88.62 $\pm$ 0.0047	87.79 $\pm$ 0.0059	87.98 $\pm$ 0.0021	87.69 $\pm$ 0.0088
	0.0001	89.12 $\pm$ 0.0028	88.69 $\pm$ 0.0041	88.27 $\pm$ 0.0038	88.61 $\pm$ 0.0036	87.93 $\pm$ 0.0040	88.24 $\pm$ 0.0024	87.53 $\pm$ 0.0042
Precision	0.0001	89.31 $\pm$ 0.0018	88.93 $\pm$ 0.0022	88.45 $\pm$ 0.0021	88.86 $\pm$ 0.0011	88.17 $\pm$ 0.0020	88.34 $\pm$ 0.0013	87.72 $\pm$ 0.0022
Recall	0.0001	89.12 $\pm$ 0.0028	88.69 $\pm$ 0.0041	88.27 $\pm$ 0.0038	88.61 $\pm$ 0.0036	87.93 $\pm$ 0.0040	88.24 $\pm$ 0.0024	87.53 $\pm$ 0.0042
F1-Score	0.0001	89.15 $\pm$ 0.0025	88.72 $\pm$ 0.0038	88.21 $\pm$ 0.0034	88.63 $\pm$ 0.0032	87.89 $\pm$ 0.0038	88.21 $\pm$ 0.0020	87.47 $\pm$ 0.0040

5.2 MNIST Digit Classification (CNN Architecture)

Table 3 summarizes performance on MNIST. SMishReLU yields a top accuracy of $99.56\% \pm 0.0002$, outperforming MishReLU (99.47%) and ReLU (99.43%).

Table 3: CNN on MNIST Dataset (Mean $\pm$ SD over 5 seeds)

Metric (%)	LR	SMishReLU	MishReLU	ReLU	Mish	ELU	LeakyReLU	SeLU
Accuracy	0.01	99.42 $\pm$ 0.0003	99.34 $\pm$ 0.0006	99.36 $\pm$ 0.0007	99.38 $\pm$ 0.0004	99.35 $\pm$ 0.0006	99.38 $\pm$ 0.0007	99.40 $\pm$ 0.0004
	0.001	99.56 $\pm$ 0.0002	99.47 $\pm$ 0.0004	99.43 $\pm$ 0.0004	99.45 $\pm$ 0.0003	99.47 $\pm$ 0.0004	99.44 $\pm$ 0.0006	99.45 $\pm$ 0.0005
	0.0001	99.43 $\pm$ 0.0002	99.38 $\pm$ 0.0002	99.37 $\pm$ 0.0003	99.37 $\pm$ 0.0004	99.34 $\pm$ 0.0002	99.36 $\pm$ 0.0005	99.25 $\pm$ 0.0004
F1-Score	0.001	99.55 $\pm$ 0.0002	99.46 $\pm$ 0.0004	99.42 $\pm$ 0.0004	99.44 $\pm$ 0.0003	99.47 $\pm$ 0.0004	99.44 $\pm$ 0.0006	99.45 $\pm$ 0.0006

5.3 CIFAR-100 Object Recognition (ResNet-18 Architecture)

On the complex CIFAR-100 dataset (Table 4), SMishReLU achieves a mean accuracy of $58.82\% \pm 0.0124$ at $LR = 0.001$, significantly outperforming MishReLU (57.65%) and ReLU (56.94%).

Table 4: ResNet-18 on CIFAR-100 Dataset (Mean $\pm$ SD over 5 seeds)

Metric (%)	LR	SMishReLU	MishReLU	ReLU	Mish	ELU	LeakyReLU	SeLU
Accuracy	0.01	47.21 $\pm$ 0.0820	45.07 $\pm$ 0.1395	40.23 $\pm$ 0.0582	52.06 $\pm$ 0.0121	49.27 $\pm$ 0.0107	49.67 $\pm$ 0.0234	37.10 $\pm$ 0.0902
	0.001	58.82 $\pm$ 0.0124	57.65 $\pm$ 0.0161	56.94 $\pm$ 0.0109	56.92 $\pm$ 0.0234	55.14 $\pm$ 0.0084	55.53 $\pm$ 0.0170	52.33 $\pm$ 0.0063
	0.0001	53.84 $\pm$ 0.0029	52.39 $\pm$ 0.0037	48.75 $\pm$ 0.0096	53.13 $\pm$ 0.0080	52.08 $\pm$ 0.0013	49.97 $\pm$ 0.0065	49.57 $\pm$ 0.0055
Precision	0.001	62.48 $\pm$ 0.0112	61.26 $\pm$ 0.0146	62.13 $\pm$ 0.0143	60.61 $\pm$ 0.0218	59.17 $\pm$ 0.0096	60.94 $\pm$ 0.0139	56.52 $\pm$ 0.0072
F1-Score	0.001	58.54 $\pm$ 0.0131	57.36 $\pm$ 0.0174	56.61 $\pm$ 0.0121	56.33 $\pm$ 0.0253	54.76 $\pm$ 0.0095	55.28 $\pm$ 0.0174	51.78 $\pm$ 0.0071

5.4 IMDB Sentiment Analysis (LSTM Architecture)

For sequence modeling on IMDB (Table 5), SMishReLU reaches $88.45\% \pm 0.0015$ accuracy, demonstrating strong generalization in recurrent cells.

Table 5: LSTM on IMDB Dataset (Mean $\pm$ SD over 5 seeds)

Metric (%)	LR	SMishReLU	MishReLU	ReLU	Mish	ELU	LeakyReLU	SeLU
Accuracy	0.01	88.45 $\pm$ 0.0015	87.92 $\pm$ 0.0021	87.56 $\pm$ 0.0026	87.68 $\pm$ 0.0012	87.52 $\pm$ 0.0028	87.60 $\pm$ 0.0030	87.31 $\pm$ 0.0032
	0.001	87.38 $\pm$ 0.0032	86.85 $\pm$ 0.0061	87.07 $\pm$ 0.0042	84.69 $\pm$ 0.0491	86.41 $\pm$ 0.0094	87.50 $\pm$ 0.0031	86.30 $\pm$ 0.0043
	0.0001	88.45 $\pm$ 0.0015	87.92 $\pm$ 0.0021	87.56 $\pm$ 0.0025	87.68 $\pm$ 0.0012	87.51 $\pm$ 0.0028	87.60 $\pm$ 0.0031	87.31 $\pm$ 0.0032

5.5 Reuters Topic Classification (BiLSTM Architecture)

On the 46-class Reuters dataset (Table 6), SMishReLU secures a top accuracy of $68.91\% \pm 0.0076$ compared to 67.67% for MishReLU.

Table 6: BiLSTM on Reuters Dataset (Mean $\pm$ SD over 5 seeds)

Metric (%)	LR	SMishReLU	MishReLU	ReLU	Mish	ELU	LeakyReLU	SeLU
Accuracy	0.01	68.91 $\pm$ 0.0076	67.67 $\pm$ 0.0098	65.34 $\pm$ 0.0143	66.22 $\pm$ 0.0161	66.87 $\pm$ 0.0109	65.22 $\pm$ 0.0157	66.17 $\pm$ 0.0165
	0.001	68.62 $\pm$ 0.0152	57.27 $\pm$ 0.0215	56.30 $\pm$ 0.0551	56.43 $\pm$ 0.0287	57.36 $\pm$ 0.0181	57.80 $\pm$ 0.0201	59.07 $\pm$ 0.0169
	0.0001	67.62 $\pm$ 0.0094	66.33 $\pm$ 0.0119	63.11 $\pm$ 0.0138	65.09 $\pm$ 0.0184	65.74 $\pm$ 0.0136	63.61 $\pm$ 0.0180	64.90 $\pm$ 0.0228

6 Conclusion

In this paper, we identified and resolved a crucial theoretical limitation in the MishReLU activation function: its first-derivative jump discontinuity at $x = 0$. We presented a literature review of 16 recent activation function publications (2020–2025) and introduced **SMishReLU**, incorporating an exact scaling factor $\alpha = \frac{1}{\tanh(\ln 2)} \approx 1.6667$ to achieve complete C^1 differentiability across the domain. Empirical evaluations across five standard vision and text classification benchmarks confirmed that SMishReLU provides superior accuracy, lower gradient variance, and faster convergence.

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